Ahmedabad University

CSE-401 Database Management System

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Project Title: Supply Chain and Logistic Management

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1. Introduction

Supply Chain and Logistics Management System is an application that is based on databases to effectively manage the flow of goods, transactions, and information between customers, distributors and products.

The system incorporates various entities like users, customers, distributors, products, orders, payments, inventory and reviews into a database. It provides smooth integration of ordering, inventory management, processing as well as sales analysis.

The project builds on the principles of the relational database, namely, entity relationships, constraints, triggers, stored procedures, or functions, to ensure data integrity and automate the major operations. As an illustration, order processing and inventory validation triggers are some of the procedures that are used to maintain consistency and reliability in the system in real time.

2. Objectives

The overall goals of this project include:

- To create a relational database structure of the supply chain management.
- To automate order handling, inventory management, and payment management.
- To maintain data integrity on a basis of constraints, triggers and relationships.
- In order to monitor customer activities (use of carts, orders, and reviews).
- To deliver analytical data such as sales and performance of the products.
- To enhance efficiency in the logistics operations through minimization of manual interaction.

3. Functional Requirements

This system must be able to sustain the following functionalities:

User & Customer Management

- Users are able to create, log in and control profiles.
- The customers are able to order, manage carts and leave reviews.
- Product and Distributor Management.
- The distributors may post products categorized and priced.
- Inventory is associated with products to monitor stocks.

Cart & Order Processing

- Customers are able to add/ delete items in the cart.
- Total order value is computed automatically in the system.
- A cart contains items that are used to create orders via a stored procedure.
- Order items are stored having quantity and price information.

Payment Management

- Each order is registered with payments.
- Successful payments initiate updates in sales history.

Inventory Management

- The inventory is updated upon every order.
- A trigger is used to ensure that the stock does not go below certain minimum levels.

Review System

- Products can be rated and reviewed by the customers.
- Functions are used to calculate average product ratings.

Sales & Reporting

- All transactions are kept in terms of sales history.
- Reports produce overall revenue and high selling products.
- The calculation of loyalty discounts depends on the expenditure of the customers.

4. Non-Functional Requirements

The following qualities should be matched by the system:

Performance

- Performance in query processes on large datasets.
- Streamlined processes of order processing and reporting.

Security

- Protective management of user credentials.
- Customer and distributor role-based access.

Scalability

- Capacity to support growing users, products and transactions.

Maintainability

- Procedural and functional modular database design.
- Simple to maintain business rules (e.g. discount logic).

Reliability

- Constraints and triggers to ensure the consistency of data.
- Blocking of invalid transactions (e.g. lack of stock).

Usability

- Easy user interaction (cart - order- payment).
- Proper organization of products and orders.

Entity- Relationship Diagram

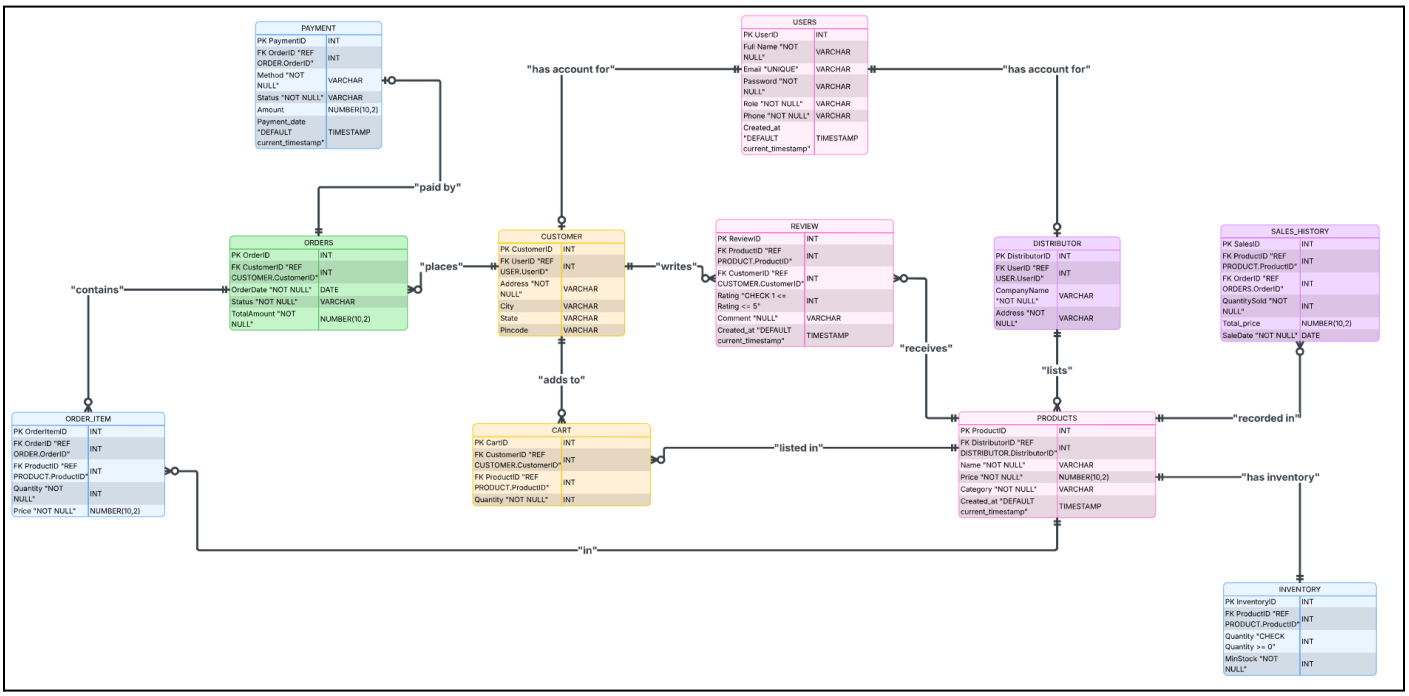


Table Design

- **USERS Table**

Field	Type	Constraint
UserID	INT	Primary Key
Full Name	VARCHAR	NOT NULL
Email	VARCHAR	UNIQUE
Password	VARCHAR	NOT NULL
Role	VARCHAR	NOT NULL
Phone	VARCHAR	NOT NULL
Created_at	TIMESTAMP	DEFAULT current_timestamp

- **DISTRIBUTOR Table**

Field	Type	Constraint
DistributorID	INT	Primary Key
UserID	INT	Foreign Key
CompanyName	VARCHAR	NOT NULL
Address	VARCHAR	NOT NULL

- **CUSTOMER Table**

Field	Type	Constraint
CustomerID	INT	Primary Key
UserID	INT	Foreign Key
Address	VARCHAR	NOT NULL

City	VARCHAR	-
State	VARCHAR	-
Pincode	VARCHAR	-

- **PRODUCTS Table**

Field	Type	Constraint
ProductID	INT	Primary Key
DistributorID	INT	Foreign Key
Name	VARCHAR	NOT NULL
Price	DECIMAL	NOT NULL
Category	VARCHAR	NOT NULL
Created_at	TIMESTAMP	DEFAULT current_timestamp

- **INVENTORY Table**

Field	Type	Constraint
InventoryID	INT	Primary Key
ProductID	INT	Foreign Key
Quantity	INT	CHECK (Quantity ≥ 0)
MinStock	INT	NOT NULL

- **ORDERS Table**

Field	Type	Constraint
OrderID	INT	Primary Key
CustomerID	INT	Foreign Key
OrderDate	DATE	NOT NULL

Status	VARCHAR	NOT NULL
TotalAmount	DECIMAL	NOT NULL

- **ORDER_ITEM Table**

Field	Type	Constraint
OrderItemID	INT	Primary Key
OrderID	INT	Foreign Key
ProductID	INT	Foreign Key
Quantity	INT	NOT NULL
Price	NUMERIC(10,2)	NOT NULL

- **PAYMENT Table**

Field	Type	Constraint
PaymentID	INT	Primary Key
OrderID	INT	Foreign Key
Method	VARCHAR	NOT NULL
Status	VARCHAR	NOT NULL
Amount	NUMERIC(10,2)	NOT NULL
Payment_date	TIMESTAMP	DEFAULT current_timestamp

- **REVIEW Table**

Field	Type	Constraint
ReviewID	INT	Primary Key
ProductID	INT	Foreign Key
CustomerID	INT	Foreign Key

Rating	INT	CHECK (Rating BETWEEN 1 AND 5)
Comment	VARCHAR	NULL
Created_at	TIMESTAMP	DEFAULT current_timestamp

- **CART Table**

Field	Type	Constraint
CartID	INT	Primary Key
CustomerID	INT	Foreign Key
ProductID	INT	Foreign Key
Quantity	INT	NOT NULL

- **SALES_HISTORY Table**

Field	Type	Constraint
SalesID	INT	Primary Key
ProductID	INT	Foreign Key
OrderID	INT	Foreign Key
QuantitySold	INT	NOT NULL
Total_price	NUMERIC(10,2)	NOT NULL

Drive Link for code for Database creation:

Database Creation.pdf

Drive Link for code to connect front-end and back-end:

Server code(To connect frontend & backend.pdf

Drive Link for code for functions:

Functions.pdf

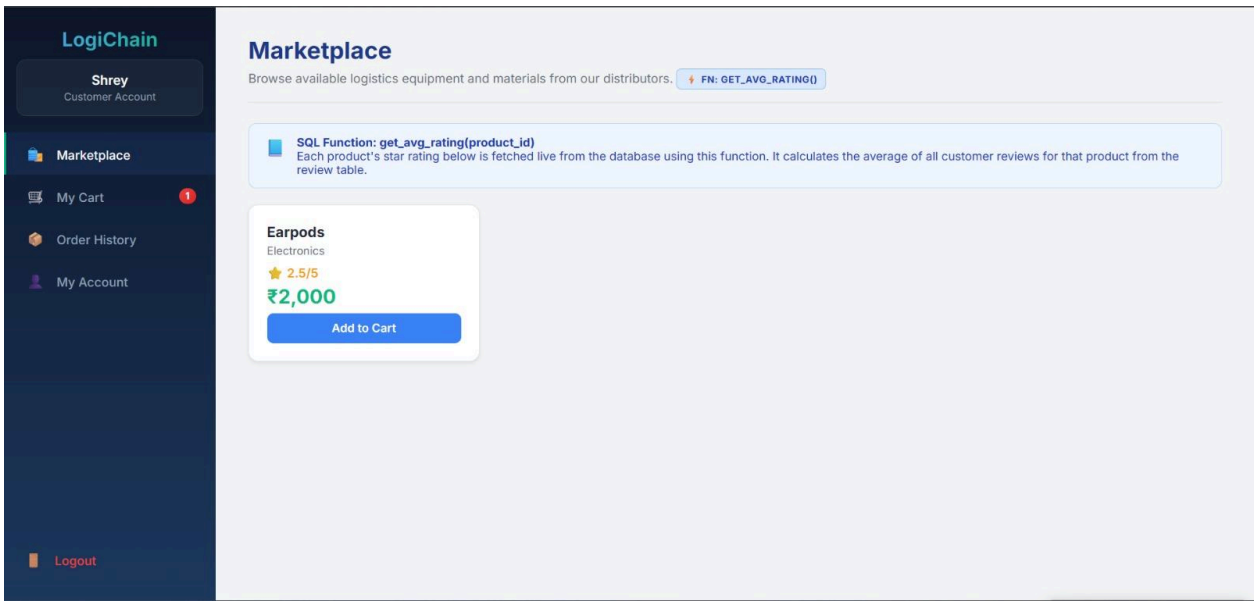
Drive Link for code for procedures:

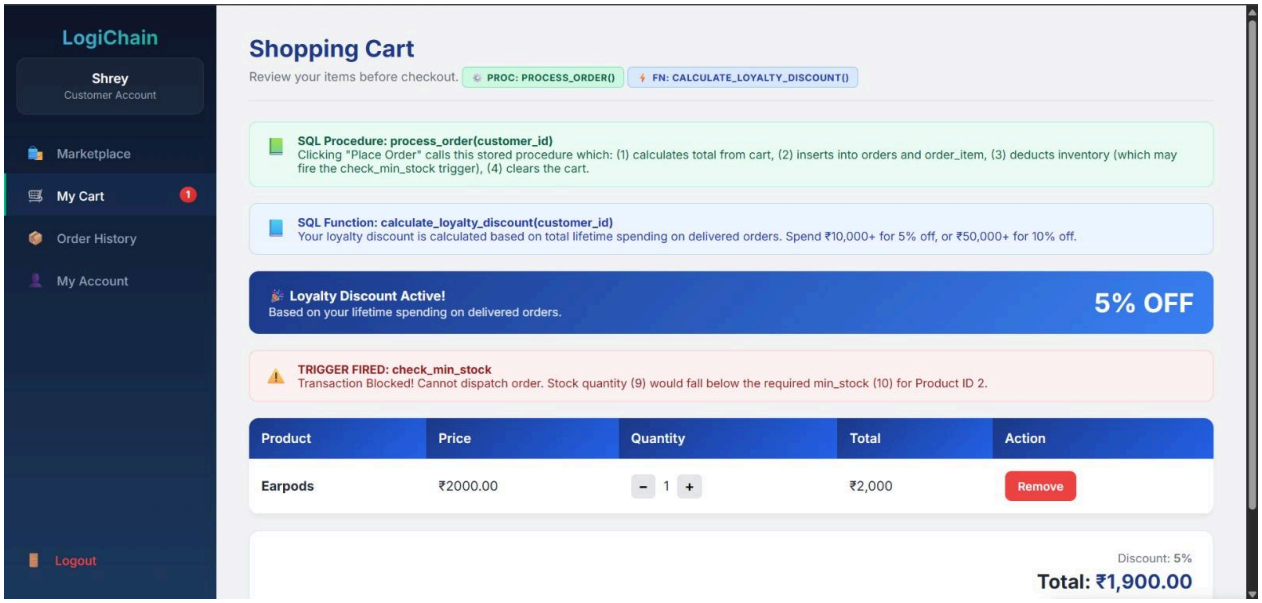
Procedures.pdf

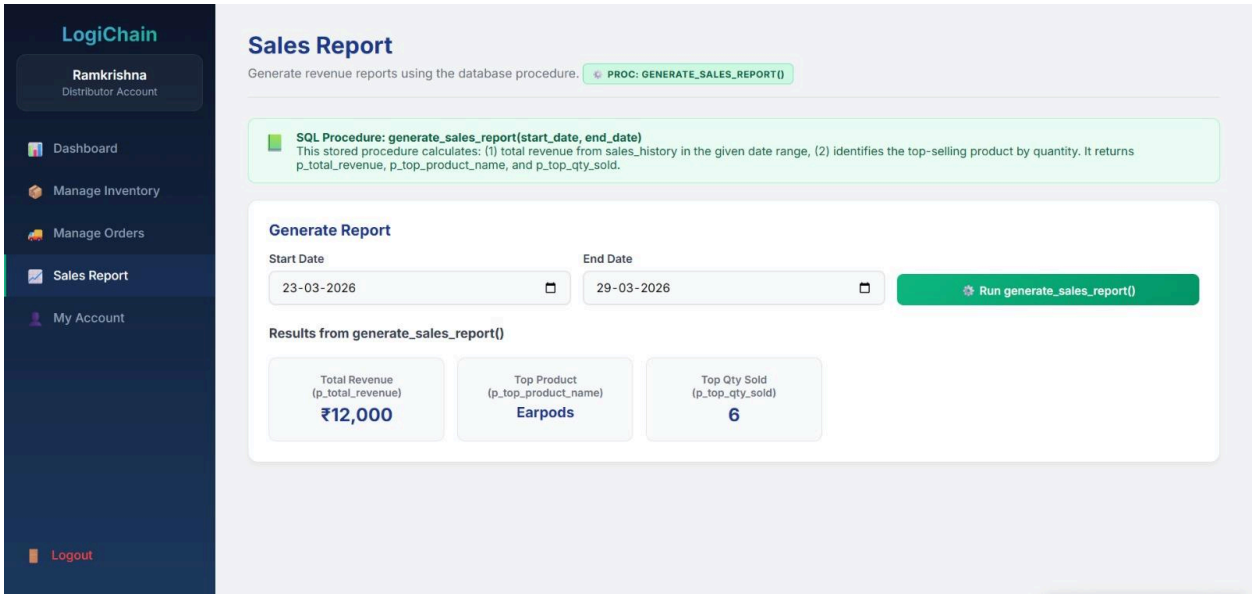
Drive Link for code for triggers:

Triggers .pdf

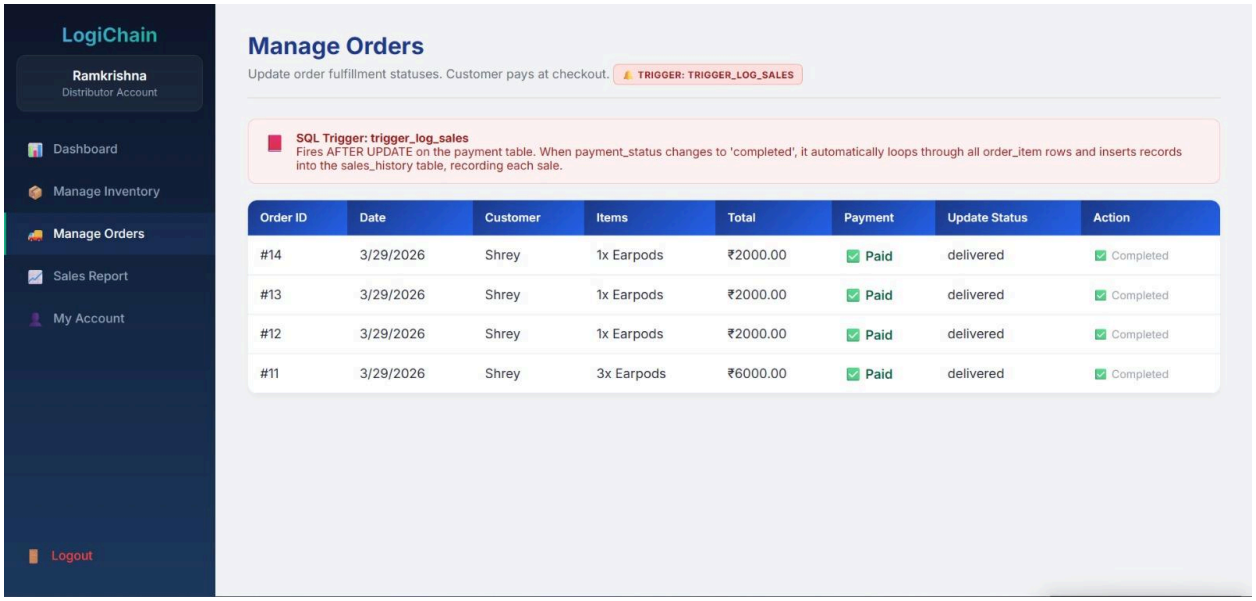
Screenshots of results generated after procedures and functions are called on the front-end

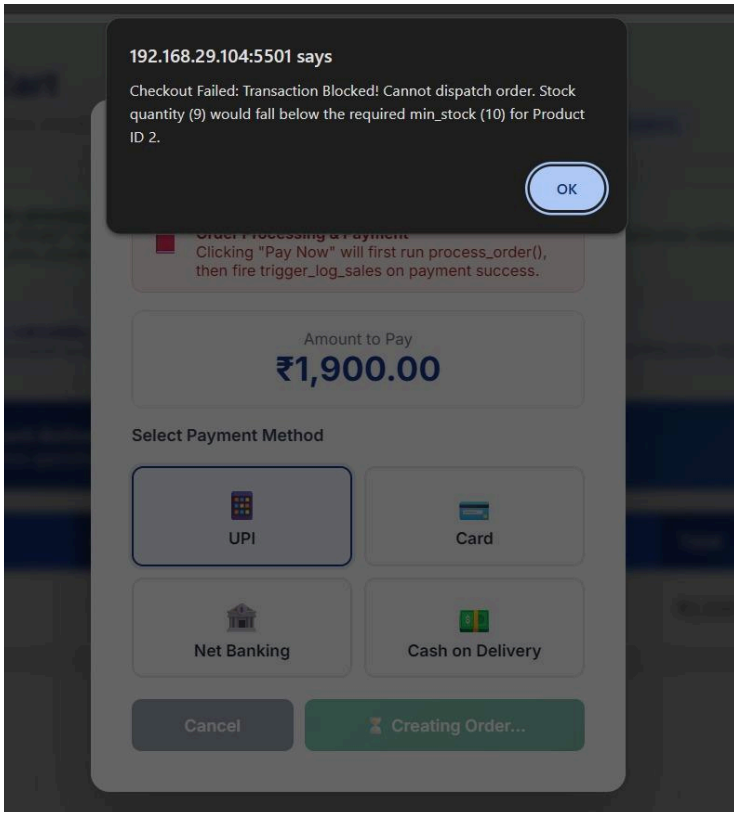


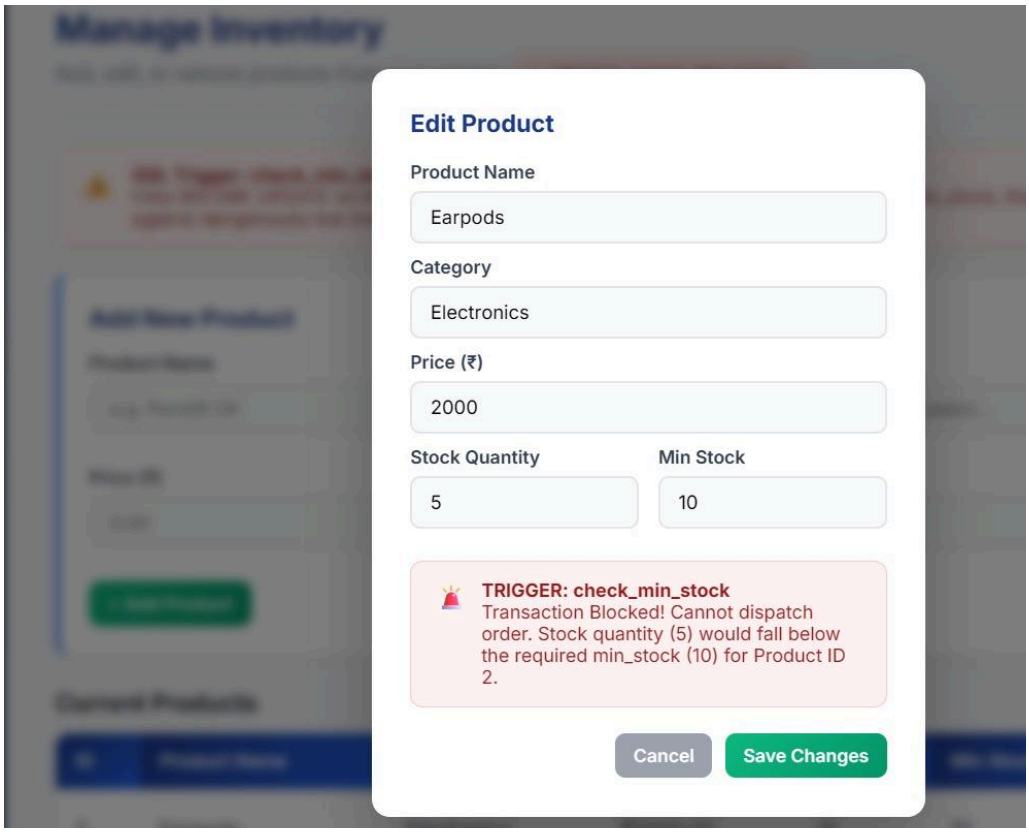




Screenshots of errors generated on the front-end when trigger is fired




192.168.29.104:5501 says
Checkout Failed: Transaction Blocked! Cannot dispatch order. Stock quantity (9) would fall below the required min_stock (10) for Product ID 2.
OK
Clicking "Pay Now" will first run process_order(), then fire trigger_log_sales on payment success.
Amount to Pay
₹1,900.00
Select Payment Method
UPI
Card
Net Banking
Cash on Delivery
Cancel
Creating Order...


Edit Product
Product Name
Earpods
Category
Electronics
Price (₹)
2000
Stock Quantity
5
Min Stock
10
TRIGGER: check_min_stock
Transaction Blocked! Cannot dispatch order. Stock quantity (5) would fall below the required min_stock (10) for Product ID 2.
Cancel
Save Changes